SURFACE CALL

Surface call
canonical / likely vs low / uncertain / no [unweighted κ] — 0.82 / 0.95

Confidence tier
no < uncertain < low < likely < canonical [weighted κ] — 0.85 / 0.92

PROTEIN CLASSIFICATION

Protein family
receptor · enzyme · transporter · miscellaneous [unweighted κ] — 1.00 / 1.00

Subcategory
single-pass T1/T2 · multi-pass · GPCR · tetraspanin · GPI · other [unweighted κ] — 0.97 / 1.00

Primary compartment
plasma membrane · endolysosome · ER · Golgi · cytosol · secreted … [unweighted κ] — 0.82 / 1.00

SURFACE EVIDENCE

Surface call reason
e.g. multipass · GPI · tissue-restricted · induced · secreted [unweighted κ] — 0.79 / 0.83

Surface accessibility
no < low < moderate < high [weighted κ] — 0.90 / 0.95

Surface specificity
mostly intracellular < mixed < surface-dominant [weighted κ] — 0.83 / 0.91

Evidence grade
weak < indirect < direct (1 method) < direct (multi) [weighted κ] — 0.83 / 0.92

Confidence
low < moderate < high [weighted κ] — 0.80 / 0.91

Live-cell surface evidence
yes · no [unweighted κ] — 0.75 / 0.90

Surface localization on overexpression
yes · no [unweighted κ] — 0.87 / 0.92

BIOLOGICAL CONTEXT

State dependence
low < moderate < high (unclear excluded, n=37) [weighted κ] — 0.72 / 0.62

Induction trigger
none · oncogenic · immune · stress · infection · other … [unweighted κ] — 0.60 / 0.74

Expression level
absent < low < moderate < high [weighted κ] — 0.85 / 0.89

Expression breadth
rare < restricted < broad < pan-tissue [weighted κ] — 0.83 / 0.98

Low endogenous expression
yes · no [unweighted κ] — 0.87 / 0.96

Tumor-associated
yes · no [unweighted κ] — 0.90 / 1.00

ACCESSIBILITY RISKS

Shed form
yes · no (rare: 4/50 yes) [unweighted κ] — 0.56 / 1.00

Secreted form
yes · no [unweighted κ] — 0.91 / 0.91

Secreted form source
proteolytic · splicing · both · unknown · n/a [unweighted κ] — 0.84 / 0.92

Epitope masking
yes · no [unweighted κ] — 0.72 / 0.84

Co-receptor dependency
none < modulatory < required (unknown excluded, n=45) [weighted κ] — 0.86 / 0.93

Restricted membrane subdomain
yes · no [unweighted κ] — 0.80 / 0.92

Restricted subdomain type
apical · basolateral · junctional · ciliary · synaptic … [unweighted κ] — 0.72 / 0.86

Known ligand
yes · no [unweighted κ] — 0.80 / 0.80

0.0 0.2 0.4 0.6 0.8 1.0

Cohen's κ vs published record

Full re-run
Fixed-evidence replay
Moderate (0.60–0.79)
Strong (0.80–0.90)
Almost perfect (>0.90)